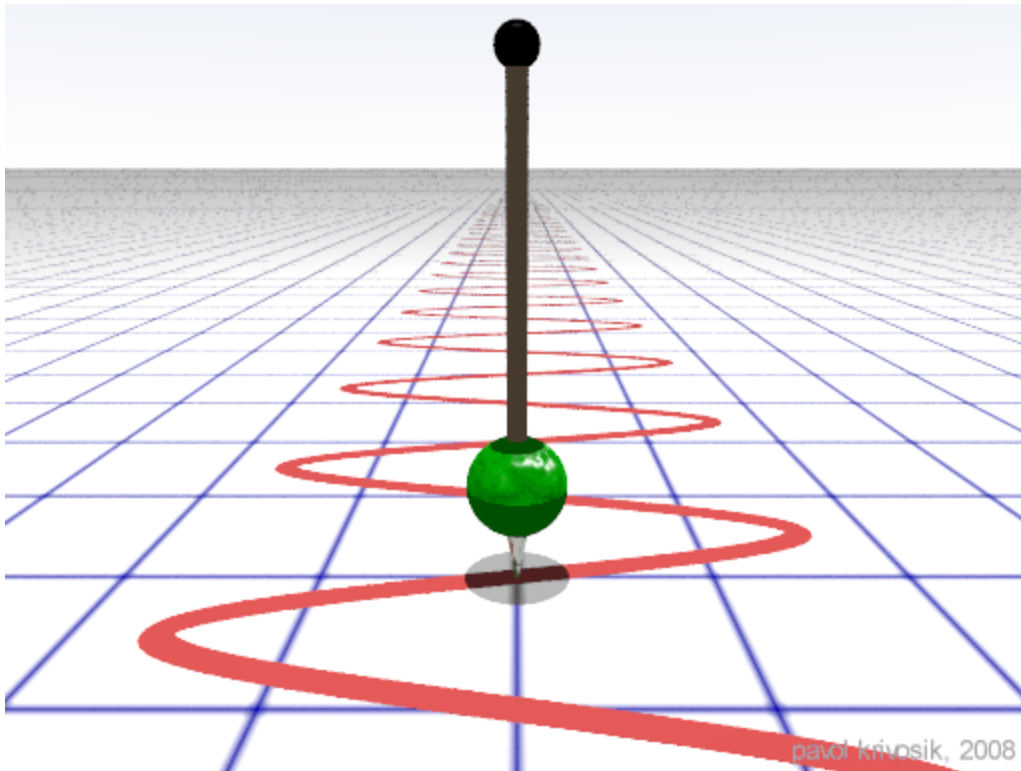


popquiz pendulum and simple harmonic motion, power, $U(r)$ ▲▼

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:39pm

Quiz Instructions



Question 1 12.5 pts

1. Which of the following is NOT a correct unit for work?

- ☐ A ft·lb
- ☐ B watt
- ☐ C newton·meter
- ☐ D joule

☐ D

☐

C

☐

A

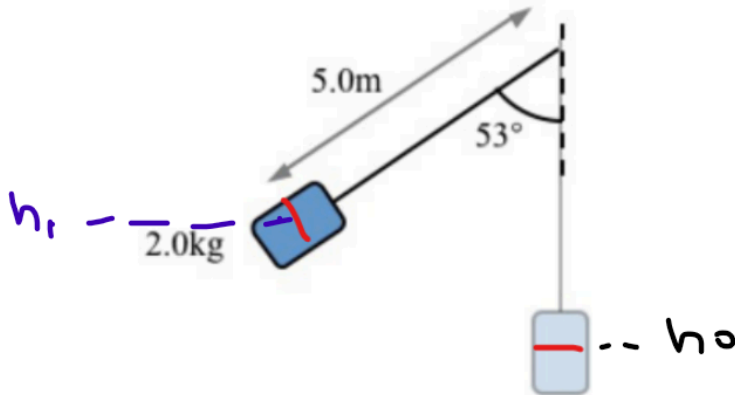
☐

B

☒

Question 2 12.5 pts

Here is a pendulum before it is released.



The red line shows the center of the blob (center of mass). 5m is the distance between axis of rotation and the red line. be how much the height has changed? use trig. Call that height $h = h_1 - h_0$

hint: its a subtraction. as engineers it should not too hard. Use trig. U need $h_1 - h_0$. Ask me at worse.

☐

about 0.5m

☐

about 2m

☐

about 2.5m

☐

about 1m

☒

Question 3 12.5 pts

Building on the previous question. The pendulum shown above is released and its PE (mgh) is turned to KE ($0.5mV^2$).

there is no non-conservative force involved (so no friction).

. What is the speed of the bob at the bottom of the swing ? (when $h=0$)).

You can Use : work done by gravity = change in kinetic energy.

(mass does not matter right? you see why?) - **work is weight x displacement . displacement is $h_1 - h_0$**

or You can also use $PE = KE$. Suppose $PE=0$ when $h = h_0$ and h was computed previously.

take $g=10\text{m/s/s}$

☐

about 60m/s

☐

about 7.7m/s

☐

about 40m/s

☐

about 6.3 m/s



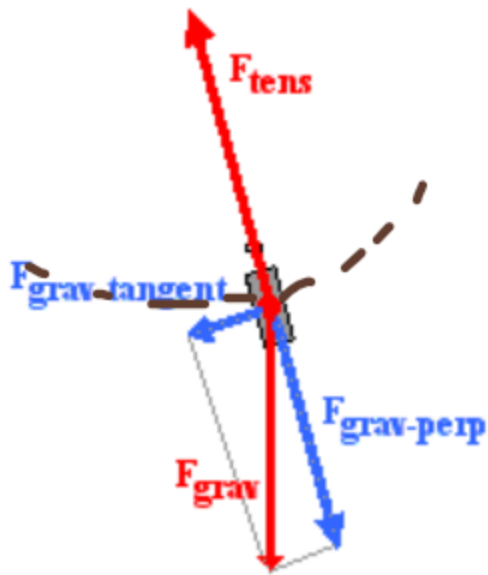
Question 4 12.5 pts

Building on the previous problem. I drew for you the free diagram for the pendulum (thank you). . There are 2 forces applied to the bob. gravity and the tension (red) What is true ? The weight has 2 components in blue here.

The restoring force is the force that tries to bring back the pendulum to its rest position (lowest position where PE is minimum and tension = weight)

What is true (refer to diagram)

Free Body Diagram (with components)


☐

the restoring force is the weight (F_g in red)

☐

The restoring force is the tangential component of the weight

☐

the restoring force is the perpendicular component of the weight

☐

the restoring force is the tension T

☐

Question 5 12.5 pts

Building on the previous question what is the expression for the restoring force?

the weight of the bob is mg



☐ $mg\sin(\alpha)$

☐ $mg\cos(\alpha)$

☐ mg

☐ $mg\tan(\alpha)$



Question 6 12.5 pts

Building on previous question. If an angle is small then $\sin(\alpha) = (\alpha)$ in radians

This is a first order approximation. Therefore the restoring force becomes

☐ inversely proportional to (α)

☐ constant. independent of the displacement

☐ proportional to (α) squared

☐ proportional to displacement (α)



Question 7 12.5 pts

Building up. When a restoring force is proportional to displacement (here angular displacement) the motion is

hint: check the front image of the popquiz

☐

the motion describes a parabola

☐

the motion describes a step function

☐

a simple harmonic motion and describes a sine wave

☐

the motion describes a zig zag



Question 8 12.5 pts

A bucket of water with a total weight of 50 Newtons is lifted at constant velocity up a 10 meter deep well. If it takes 20 seconds to raise the bucket this distance, the average Power required to lift the bucket is

(there is an equation for power when you have the speed see my slides or equation sheet)

☐

25J

☐

500W

☐

2.5W

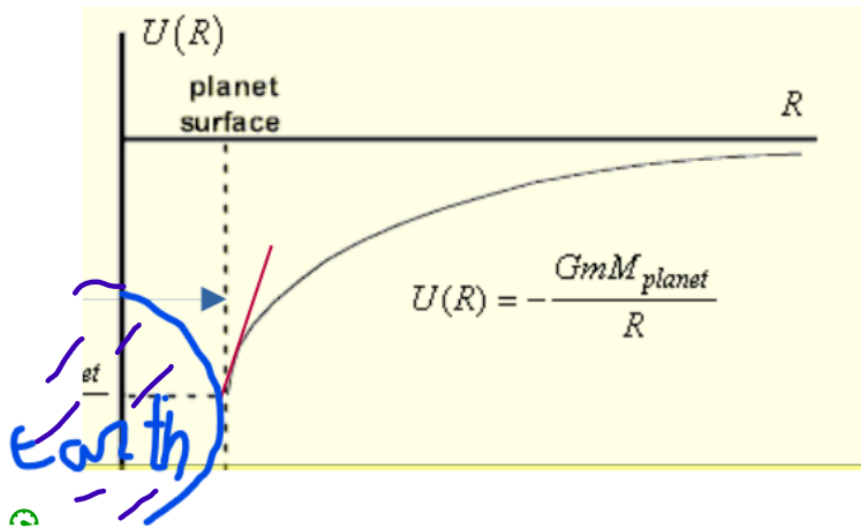
☐

25W



Question 9 5 pts

Bonus question



This graph shows the gravitational potential energy of planet Earth. You can see the Earth's radius. (planet surface). R is the distance from Earth center.

What is true:

- ☐ U is negative, U follows an inverse relationship and is 0 at infinity
- ☐ close to Earth U becomes positive
- ☐ U is negative and follows an exponential. it is 0 at infinity
- ☐ U is negative and follows an inverse square relationship and is 0 when $r=\text{infinity}$

Not saved

Submit Quiz